

Technology Stack

Date	1 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

In this project, **the technology stack includes HTML, CSS, Bootstrap, and JavaScript for building the user interface, Python and Flask for backend development, Scikit-learn and XGBoost for developing machine learning models, Pandas and NumPy for data preprocessing, Pickle for model serialization, Git and GitHub for version control, and Google Colab/Jupyter Notebook for data analysis, experimentation, and model training. Together, these technologies enable the development of an accurate, reliable, and user-friendly credit card approval prediction system.**

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, JavaScript	Provides a responsive and user-friendly interface for entering applicant details and displaying prediction results.
2	Backend / Server-Side	Python, Flask	Handles user requests, processes input data, loads the trained machine learning model, and returns prediction results efficiently.
3	Machine Learning	Scikit-learn, XGBoost	Used to train, evaluate, and compare multiple classification algorithms for accurate credit card approval prediction.
4	Data Processing	Pandas, NumPy	Performs data cleaning, preprocessing, feature engineering, and numerical computations before model training and prediction.
5	Model Serialization	Pickle	Saves the trained machine learning model as a .pkl file for use in the Flask application.
6	Version Control	Git, GitHub	Tracks source code changes, manages project versions, and enables collaboration and code backup.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
7	Notebook Environment	Google Colab, Jupyter Notebook	Used for exploratory data analysis, feature engineering, model development, experimentation, and training.